

Placement of the end-of-proof mark

Lemma 1. *For every $n \geq 1$ we have $1 + 2 + \cdots + n = n(n+1)/2$.*

Proof of Lemma 1. Write the sum twice, once forwards and once backwards, and add the two rows term by term. Each of the n columns sums to $n+1$, hence

$$2 \sum_{k=1}^n k = n(n+1). \quad \square$$

Theorem 2. *For every $n \geq 1$, $\sum_{k=1}^n (2k-1) = n^2$.*

Proof. By Lemma 1,

$$\begin{aligned} \sum_{k=1}^n (2k-1) &= 2 \sum_{k=1}^n k - n \\ &= n(n+1) - n \\ &= n^2. \end{aligned} \quad \square$$

Theorem 3. *Let a, b be integers with $a \mid b$ and $b \mid a$. Then $a = \pm b$.*

Proof. We treat two cases.

1. If $a = 0$ then $b = 0$ as well, since 0 divides only itself.
2. If $a \neq 0$, write $b = ka$ and $a = lb$. Then $a = lka$, so $lk = 1$ and $k = \pm 1$. $\square$

Theorem 4. *There are infinitely many primes.*

Proof. Suppose $p_1, \dots, p_r$ are all the primes and let $N = p_1 \cdots p_r + 1$.

- No p_i divides N , because it leaves remainder 1.
- But $N > 1$ has a prime factor, which is a contradiction. $\square$

Second proof. The series $\sum_p 1/p$ diverges, which is impossible for a finite set of primes. We only sketch this argument here. $\square$

Theorem 5. *The number $\sqrt{2}$ is irrational.*

Proof. If $\sqrt{2} = p/q$ in lowest terms, then $p^2 = 2q^2$ is even, so p is even, so q is even, contradicting lowest terms. $\blacksquare$

Proof with a displayed equation. Equivalently, no positive integers p, q satisfy

$$p^2 = 2q^2 \tag{1}$$

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A remark in proof form. The same argument shows that $\sqrt{m}$ is irrational whenever m is not a perfect square. QED